

Effective Sample Generation for Industrial Control Systems: A GAN-Based Approach

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Abstract—In industrial control systems (ICSs), safeguarding system security is vital, and intrusion detection is an effective protective measure. In real-world ICS environments, however, the training data for intrusion-detection models are dominated by normal traffic, with very few abnormal samples. This scarcity prevents the model from fully learning abnormal characteristics, thereby limiting anomaly-detection performance. ICS traffic is also noisy and often captured incompletely, which is especially problematic because it possesses strong temporal dependencies. Moreover, ICS operations tend to be highly regular and follow a strict logical order, causing data to cluster densely in certain regions while edge regions remain sparsely populated—an imbalance that easily leads to misclassification. To tackle data imbalance, missing temporal features, and edge-sample scarcity, we propose the DualCGAN-AE approach. It employs two conditional generative adversarial networks to perform dynamic learning on global and edge features and to enrich samples with temporal information; a multimodel discriminator is incorporated to enhance training stability. We validated the method on two datasets, assessing both the quality of the generated samples and their impact on intrusion detection performance. Experimental results show that the samples produced by DualCGAN-AE significantly boost detection accuracy.

Index Terms—Edge distribution, generative adversarial network (GAN), industrial control system (ICS), intrusion detection.

I. INTRODUCTION

WITH the ongoing development of industrial control systems (ICSs), their scale and complexity are continuously increasing, as seen in natural gas plants, power plants, and water treatment facilities. Within these systems, devices, such as sensors, follow instructions from upper level components, such as programmable logic controllers (PLCs) and remote terminal units. They carry out periodic operations based on logical patterns and transmit control commands via network traffic to the supervisory control and data acquisition (SCADA) control

center for monitoring and management. This comprehensive framework enables effective monitoring of industrial processes, facilitating timely anomaly detection [1]. However, as ICS advances, malicious attacks on sensors or higher level machines are also on the rise. Consequently, real-time intrusion detection for time-series traffic data has become a research priority, and improving detection accuracy by leveraging the temporal nature of industrial traffic data remains a long-term challenge [2].

At present, numerous remarkable research outcomes exist in the realm of ICS intrusion detection, with machine learning and deep learning approaches being predominant. These techniques analyze ICS traffic, compare key fields in normal traffic, extract crucial semantic features from ICS traffic, and employ these features for anomaly detection [3]. Thanks to their capacity for autonomous learning and enhancement, these methods have somewhat reduced the costs associated with feature selection. As a result, recent research has predominantly focused on this area.

However, as new attack types continue to emerge, legacy models gradually lose their effectiveness and struggle to counter these novel threats. Consequently, leveraging unsupervised learning for ICS intrusion detection is vital for defending against an increasingly diverse array of attacks [4]. Yet, because attack samples are scarce, the training data consist almost entirely of normal traffic, resulting in severe class imbalance. This imbalance readily leads to overfitting during model training, thereby degrading the model's decision accuracy. Moreover, the ICS environment is inherently noisy, often corrupting traffic data and even causing partial data loss—situations to which the logic-driven operations of ICS are especially sensitive [5]. Therefore, completing missing traffic data and mitigating the impact of heavy noise on model judgments are crucial for improving training precision, with data repair and augmentation playing key roles in this process.

To address the issues in industrial traffic sample generation, this article proposes the DualCGAN-AE sample generation approach, with specific contributions as follows.

- 1) We propose an effective sample generation approach that employs two generative adversarial networks (GANs) in an interactive training scheme. The first GAN learns the overall data distribution; its low-scoring samples are passed to the second GAN as learning targets. This allows the second GAN to focus on the data's edge-distribution characteristics and generate samples with those properties, thereby mitigating the long-tail distribution problem in the dataset.

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- 2) To enhance training stability and cope with potential temporal data loss in industrial traffic samples, we introduce a time interval matrix (TIM) to preserve the completeness of temporal features. In addition, we adopt a multimodel discriminator: the single discriminator score is replaced with the weighted average of multiple model scores, which stabilizes the discriminator's training process.
- 3) To resolve the issue of label-containing sample generation and enable category-specific sample generation, modify the original GAN network to a conditional GAN (CGAN). Introduce label condition information to achieve controllable and targeted data generation, while still meeting the training requirements of unsupervised learning for labeled data.
- 4) Experiment on two different datasets and meticulously evaluate the generated samples. Assess our proposal in terms of data quality and its impact on intrusion detection performance.

The rest of this article is organized as follows. Section II reviews related work. Section III details our method. Section IV presents results and discussion. Finally, Section V concludes this article.

II. RELATED WORKS

In recent years, research on addressing sample data imbalance has fallen into two categories: traditional artificial data generation techniques and GAN-based data augmentation methods. Traditional artificial data generation techniques rebalance datasets by increasing the number of minority class samples. Common approaches, such as random oversampling and the synthetic minority oversampling technique (SMOTE) algorithm, raise the minority class's share by duplicating or synthetically creating samples. Oversampling, however, can introduce overfitting, whereas undersampling risks discarding critical information. GAN-based data-augmentation methods use generative models to create new minority-class samples from random noise, producing more diverse data that better match the original distribution. CGANs incorporate auxiliary information, such as class labels during generation, yielding samples that are both more structured and more varied. These GAN-based approaches have proven highly effective for remedying sample imbalance, particularly with complex data types, such as images and text.

SMOTE is a common sample generation solution that alleviates data imbalance to some extent. Dablain et al. [6] proposed DeepSMOTE, a method specifically designed for deep learning models. It preserves the attributes of original images during processing and generates high-quality artificial images. Composed of an encoder or decoder framework, SMOTE-based oversampling, and a penalty-enhanced loss function, DeepSMOTE has the advantage of not requiring a discriminator, and the generated images are information-rich and suitable for visual inspection. Maldonado et al. [7] introduced a SMOTE-like method using weighted Minkowski distance to define the neighborhood of each minority-class sample. This approach better defines neighborhoods by prioritizing features relevant to the classification task and performs feature selection by discarding attributes

with weights below a threshold, addressing class overlap and centrality issues without increasing methodological complexity. Bao and Yang [8] proposed methods to tackle this issue. They pointed out that the imbalanced classification problem is a significant challenge in neural networks and machine learning. Although SMOTE is widely used to address this issue, it has a major limitation: noisy samples may be involved in the sample-synthesis process, reducing the rationality of newly synthesized samples and degrading network classification performance. Arafa et al. [9] presented RN-SMOTE. In data balancing research, random over-sampling (ROS), random under-sampling (RUS), and SMOTE [10], [11], [12] remain primary methods. RN-SMOTE introduces noisy oversampling in the minority class, uses DBSCAN for noise detection and removal, combines the processed instances with original data, and then applies SMOTE again before introducing the dataset to the underlying classifier, thereby improving evaluation performance metrics.

Combining SMOTE with various networks is an effective way to enhance its performance. For example, Liaw et al. [13] proposed the H-SMOTE-FAM model, which integrates histograms, SMOTE, and fuzzy ARTMAP (FAM) networks. This model first organizes minority-class samples into histogram bins, calculates the target number of samples for each bin, and then generates synthetic samples within each bin using SMOTE. This systematic approach addresses SMOTE's limitations in handling intraclass imbalance and small disjunctive samples. The FAM network then processes the increased data volume through incremental learning to improve classification performance. Experimental results show that H-SMOTE-FAM outperforms other SMOTE variants and GAN-based oversampling methods. Similarly, Liu et al. [14] combined SMOTE with a deep twin multihead self-attention network to develop a feature-level data augmentation method. This method uses the network structure to map original data to a feature space with better class separability and then generates samples in this space using SMOTE. This approach has demonstrated good performance on multiple public datasets. In summary, by addressing SMOTE's limitations and improving data augmentation, combining it with different networks can enhance its performance, and these combined methods have shown promising results in various applications.

In recent years, the advancements in GANs have been remarkable, leading to significant achievements in fields, such as natural language processing and image generation. Using GANs for data generation to enhance data quality and address data imbalance issues has also become one of the current research priorities. For example, Yuan et al. [15] proposed the B-GAN method for intrusion detection systems in ICS. This method captures data features through long short-term memory (LSTM) networks and uses GANs to generate high-quality abnormal samples, effectively tackling data imbalance. Chen et al. [16] introduced the DAGAN method, which constructs a dual GAN model to learn the underlying data distribution without abnormal samples. By optimizing abnormal scores based on the learned normal and marginal distributions, this approach reduces the misjudgment of marginal samples as outliers. In summary, these GAN-based methods have demonstrated great potential in solving data imbalance problems, particularly in generating

high-quality and diverse samples to enhance model performance in tasks such as intrusion detection.

Meslet-Millet et al. [17] proposed NeCSTGen, a novel architecture based on variational autoencoders (VAEs) and recurrent neural networks (RNNs) for generating diverse network traffic. This architecture can generate packets, flows, aggregates, and external data without protocol or dynamic knowledge. Its core methods include: using VAE for feature vector embedding of packets, Gaussian mixture models for similarity identification and clustering, and RNN with transition matrices for flow and aggregate generation. Li et al. [18] introduced two models, DCCSO-Res-BIGRU and GFDA-WGAN. DCCSO-Res-BIGRU extracts features through attribute core matrices and their associative relationships, while GFDA-WGAN detects imbalanced attack traffic using a conditional Wasserstein generative adversarial network (WGAN) to generate pseudosample features of unseen classes, optimized by the GFDA strategy. Geng et al. [19] proposed a few-shot anomaly-driven generation method, which leverages only a small number of real anomaly samples to guide a diffusion model in generating realistic and diverse anomalies, thereby facilitating the training of anomaly detection models. Specifically, it learns the anomaly distribution from limited real anomalies, injects the learned knowledge into an embedding vector, and then uses this vector together with a given bounding box to guide the diffusion model in generating realistic and diverse anomalies.

III. PROPOSED APPROACH

This section describes the methodology of the GAN model. By modifying the original GAN structure, introducing variant GAN models, and enhancing training stability and completeness, the model better adapts to training tasks with labeled data. It addresses issues, such as long-tail distribution and temporal data loss in industrial control traffic data, improving training effectiveness [20], sample diversity, and the generalization ability for industrial control traffic sample generation. The proposal is presented in the following three sections. Section III-A shows an overview of the overall architecture. Section III-B gives an introduction to the enhanced CGAN and stability optimization strategies. Section III-C discusses design principles and strategies of the DualCGAN-AE model.

A. Architecture of Our Scheme

To generate high-quality samples for ICS traffic, we optimize the model design using a CGAN-based approach, considering the differences between industrial and general traffic. The specific challenges are as follows.

- 1) *Imbalanced Data Distribution*: ICS traffic contains a high proportion of normal samples and few abnormal samples, a characteristic not reflected in public datasets. This leads to poor performance of models trained on public data in real ICS scenarios. The model must accurately generate abnormal samples from a large number of normal samples.
- 2) *Gaussian Distribution of Data*: Devices, such as PLCs in ICS, operate according to logical rules, causing data

to repeat and concentrate in specific regions, with sparse samples in marginal areas. This makes model training in marginal regions error-prone, necessitating supplementary data in these areas to improve accuracy.

- 3) *Temporal Characteristics and Data Integrity*: ICS traffic is highly temporal, but the complex ICS environment introduces high-noise operational data, inevitably causing traffic data loss and disrupting temporal integrity. Therefore, missing temporal traffic data must be supplemented to ensure feature completeness.

To address these issues, we propose the DualCGAN-AE model for high-quality sample generation. As shown in Fig. 1, the model consists of an encoder, a generator, and a discriminator. *Generator*: It uses a TIM to supplement missing temporal features and modifies the traditional GAN to a CGAN to handle insufficient labeled samples. *Discriminator*: It employs a multimodel weighted averaging mechanism to enhance training stability. *Dual CGAN structure*: It replaces single CGAN training to learn data features of both concentrated and marginal distributions, mitigating the long-tail distribution problem.

In DualCGAN-AE, the encoder of the autoencoder (AE) is shared by two CGAN models to ensure consistent low-dimensional latent representations of input noise and labels [21]. This design avoids initialization differences, provides rich features for the generator, and facilitates rapid learning of data distributions to improve generation quality. The encoder's effective learning of data representations helps the generator quickly capture data distributions, enhancing generation performance and alleviating mode collapse. By deeply analyzing the differences between ICS traffic and general traffic, we optimize the model design to significantly improve sample generation quality.

B. Enhanced CGAN and Stability Optimization Strategy

1) *CGAN Model*: When training generators and discriminators using traditional GAN models, the primary reason for generating low-quality fake data is the lack of effective information. To address this issue, CGANs [22], [23] have been proposed. CGANs assist adversarial training by adding extra effective information (such as original data labels and different data modes) to the generator or discriminator. Specifically, the model inputs real labels to both the generator and the discriminator, making the discriminator's output a conditional probability based on real labels and ensuring that the fake samples generated by the generator "know" their corresponding real labels [24]. The loss function of CGANs is improved based on the original GAN, as shown in the following equation:

$$\begin{aligned} \min_G \max_D V'(D, G) \\ = E_{x \sim P_{\text{data}(x)}} [\log(D(x|y))] \\ + E_{z \sim P_{z(z)}} [\log(1 - D(G(z|y)))]. \end{aligned} \quad (1)$$

In this way, CGANs use original data labels as conditional inputs to influence the generator and discriminator. This not only addresses the issue of insufficient input data in labeled imbalanced datasets but also provides insights for the architecture proposed in this article for training marginal distribution

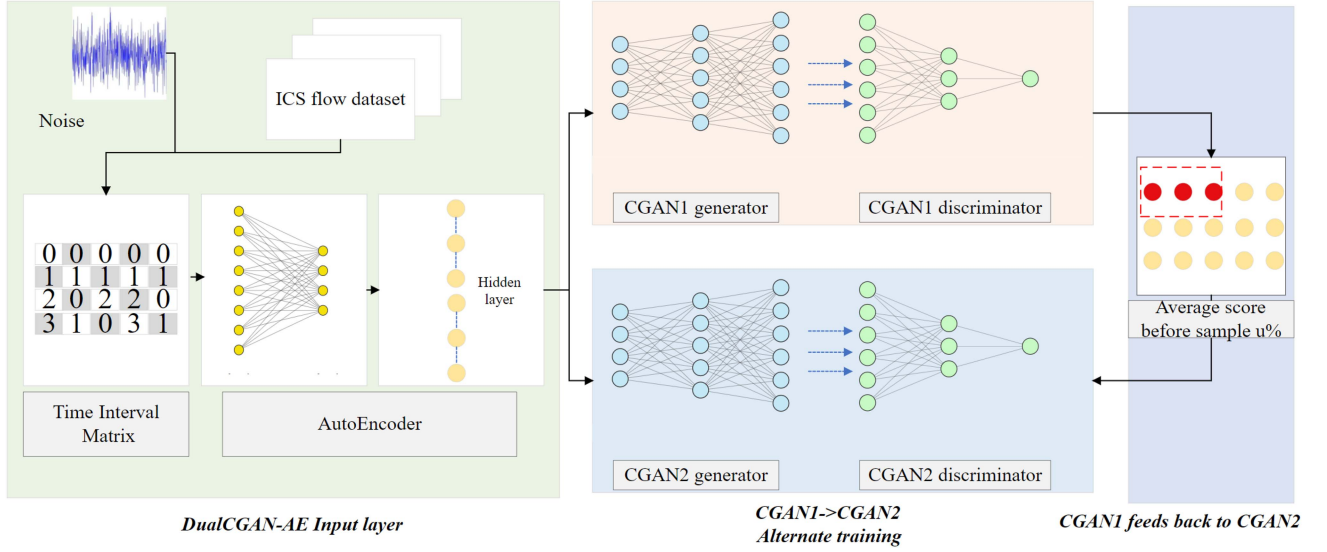


Fig. 1. DualCGAN-AE model structure.

data using dual GANs. In subsequent related research, the loss function of CGANs for marginal distribution training will be adjusted according to modified training objectives. In summary, modifying traditional GAN models into label-embedded CGAN models can alleviate data imbalance in industrial control traffic to some extent, but further enhancement of CGAN models is still needed to ensure training stability and generate higher quality data.

2) *Multimodel Discriminator*: To enhance the stability of CGAN training and avoid discriminator fitting issues caused by model differences, a multimodel discriminator is adopted instead of the traditional single-model discriminator. Specifically, three discriminator models—multilayer perceptron, gate recurrent unit (GRU), and LSTM—are used to capture data features from different perspectives through their complementarity and diversity, thereby improving overall discriminative capability. Considering the temporal characteristics of industrial control traffic data, GRU and LSTM are selected as discriminator models. These models are trained using real and generated samples to score the data, and the final discriminative result is determined by the weighted average of the three models' scores, enhancing judgment accuracy and reducing misjudgment.

During training, higher weights (40% each) are assigned to the temporal feature-capturing discriminators (GRU and LSTM), with the multilayer perceptron accounting for the remaining 20%. This strategy improves the feedback accuracy of discriminator training, enhances its generalization ability, and motivates the generator to produce traffic sample data with richer features. The overall discriminator loss function is shown in the following equation:

$$L_{\text{vote}} = \sum_{i=1}^N w_i L_i. \quad (2)$$

Here, L_i denotes the loss of the i th discriminator model, and w_i represents its corresponding weight. Although each GAN

contains three distinct discriminator models, they collectively form a traditional “generator-discriminator” structure. This improved architecture stabilizes the training process of the discriminators, diversifies the samples output by the generator, and thus meets the task objectives. Fig. 2 compares the multimodel discriminator structure used in this article with the traditional structure.

3) *Time Interval Matrix*: This article proposes the TIM to enhance model training. Before constructing TIM, it is crucial to ensure that the dataset contains time feature values suitable for timestamps. For each time-series data point denoted as s_t , we first check whether the time intervals between samples in the dataset are regular. For each missing value, we calculate the time difference between the two nearest observations; if the missing value is at the start or end of the sequence (with no preceding or succeeding observations), we use the minimum time interval in the dataset as the default. Based on these calculated time differences, we construct a matrix X with the same shape as the original time-series data, where each element x_t^d in matrix X represents the time interval corresponding to a specific timestamp.

During model training, TIM is incorporated as part of the input to the generator, helping the model better capture the temporal characteristics of the data and improve its predictive and generalization capabilities for time-series data. As shown in Fig. 3, TIM adjusts the generator's sample-generation ability by serving as part of its input. In the final DualAE-CGAN model, noise and label attributes are part of a common embedding, while TIM exists independently, acting solely as an imputation matrix for capturing time dependencies.

C. DualCGAN-AE Scheme Introduction

Based on the traditional GAN, this article introduces a second GAN model dedicated to learning marginal distribution data. By analyzing the discriminator scores from the original GAN,

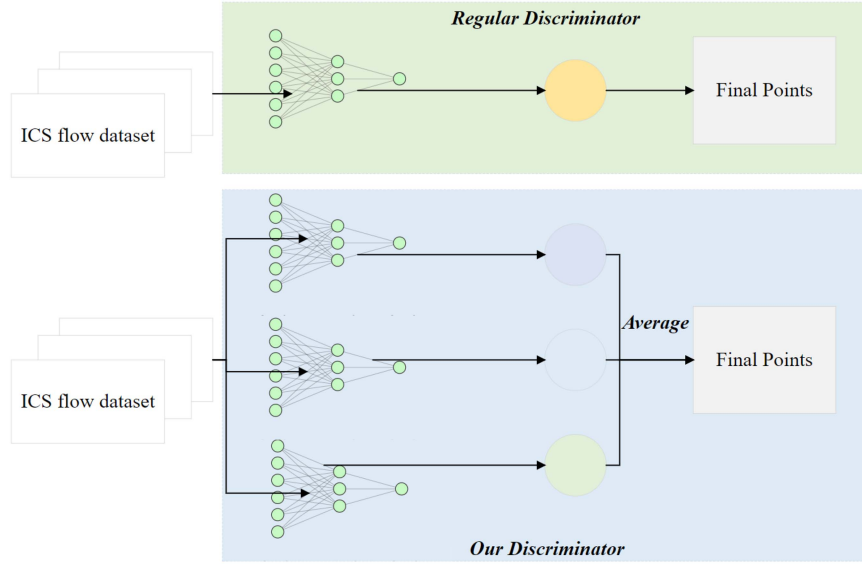


Fig. 2. Multimodel discriminator compared with traditional discriminator.

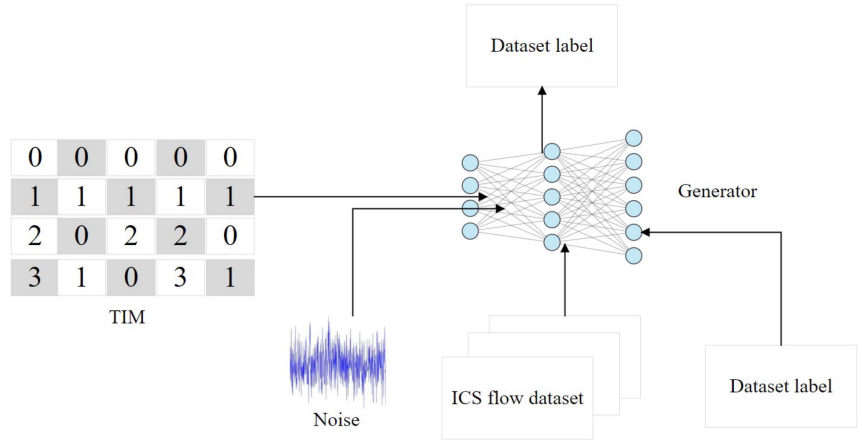


Fig. 3. TIM structure.

marginal and concentrated data in the samples are distinguished. After each training round, the discriminator scores all samples, with the scores reflecting the type of data distribution. For labeled samples, the traditional GAN is switched to a CGAN model. The proposed DualCGAN model consists of two CGANs with different training objectives:

The first CGAN (CGAN1) serves as the core model, learning features from the entire dataset through preset multiple training rounds. After each round, CGAN1 scores the data, and its results influence the second CGAN responsible for processing marginal data. Although CGAN1 cannot accurately learn marginal data features, its training process still positively impacts the final generated data due to the majority of the data being concentrated. The discriminator loss function of CGAN1 remains unchanged, as shown in the following equation:

$$L_D^{\text{CGAN1}} = -E_{x \sim P_{\text{data}}(x)}[\log(D_1(x))] - E_{z \sim P_z(z)}[\log(1 - D_1(G_1(z)))]. \quad (3)$$

The second CGAN (CGAN2) is crucial for addressing CGAN1's limitation in learning marginal sample features. During training, CGAN1 uses a training set of size N . After each training round, its discriminator scores the N samples and ranks them in ascending order. Samples with the lowest scores are selected as marginal distribution samples according to a critical ratio, and their average score $\hat{x}$ becomes the target score for CGAN2. Through alternating training of CGAN1 and CGAN2, dynamic learning of central and marginal distribution features is achieved. The proportion of marginal distribution samples is set based on industrial control traffic distribution—similar to a Gaussian distribution, which is nonuniform due to production environment noise. The Gaussian distribution formula is shown in the following equation:

$$g(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}. \quad (4)$$

The mean is μ , and the variance is σ^2 . The distribution probability within the interval $[\mu - 3\sigma, \mu + 3\sigma]$ is 99.7%, and

within $[\mu - 2\sigma, \mu + 2\sigma]$, it is 95.4%. When selecting data, two tradeoffs are considered: avoiding insufficient data points that lead to incomplete marginal distribution learning, and preventing excessive data from including central distribution samples, which could affect CGAN2 training. Therefore, in the experiment, the 95.4% probability value corresponding to the marginal distribution is used as the basis—after each CGAN1 training round, all sample scores are sorted in ascending order, and the lowest 4.6% are designated as marginal samples. Their average score, $\hat{x}$, serves as the target score for CGAN2.

In traditional GAN discriminator training, scores range from 0 to 1, where values closer to 1 indicate more realistic samples. However, in this article, both the target score and loss function for CGAN2 training are adjusted: samples closer to the target score $\hat{x}$ are more likely to be identified as marginal distribution samples. Consequently, the generative adversarial loss in the generator's loss function is adjusted as shown in (5), while the discriminator's loss function is presented in (6)

$$L_{\text{adv}}^{\text{CGAN2}} = E_{x \sim P_{\text{data}}} [\hat{x} - D_2(G_2(z))] \quad (5)$$

$$L_D^{\text{CGAN2}} = -E_{x \sim P_{\text{data}(x)}} [\log D_2(x)] - E_{z \sim P_{z(z)}} [\log (\hat{x} - D_2(G_2(z)))]. \quad (6)$$

In the DualCGAN-AE model, the encoder of the AE serves as a shared module for the two CGAN models, ensuring that input noise and labels produce consistent low-dimensional latent representations. This design avoids initialization discrepancies and provides rich features for the generator, facilitating rapid learning of data distributions and enhancing data generation quality. These improvements mean that the CGAN network includes the AE encoder in addition to the generator and discriminator. Consequently, the training loss functions of the two components have also changed: an encoder loss term is added to the original generator loss function. As shown in (7), the generator loss function of CGAN1 includes both encoder loss and adversarial loss, while the generator loss function of CGAN2 differs only in the adversarial loss component

$$\begin{aligned} L_G^{\text{CGAN2}} &= w_{\text{enc}} L_{\text{enc}}^{\text{CGAN2}} + w_{\text{adv}} L_{\text{adv}}^{\text{CGAN2}} \\ &= w_{\text{enc}} \frac{1}{N} \sum_{i=1}^N (G_E(x) - E(x'))^2 \\ &\quad + w_{\text{adv}} E_{z \sim P_{z(z)}} [\log (\hat{x} - D_2(G_2(z)))]. \end{aligned} \quad (7)$$

The generator's loss function in the DualCGAN-AE model incorporates both encoder and adversarial losses, with w_{enc} and w_{adv} representing their respective weights. The original input data are denoted by x , and the reconstructed output data are represented by x' . After the generator produces samples, the data are fed into the discriminator, which also experiences loss function changes. Unlike traditional single-model discriminators, this article employs a multimodel discriminator. Consequently, the discriminator's loss functions for CGAN2 are modified, as shown in (8).

In the DualCGAN-AE model, the generator's loss function incorporates both encoder loss and adversarial loss, where w_{enc} and w_{adv} denote their respective weights. The original input

data are denoted as x , and the reconstructed output data are represented as x' . After the generator produces samples, the data are fed into the discriminator, and the discriminator's loss function also undergoes changes. Unlike traditional single-model discriminators, this article employs a multimodel discriminator; therefore, the discriminator loss function of CGAN2 is adjusted as shown in the following equation:

$$\begin{aligned} L_D^{\text{CGAN2}} &= -E_{x \sim P_{\text{data}(x)}} [\log D_2(x)] \\ &\quad - E_{z \sim P_{z(z)}} [\log (\hat{x} - D_2(G_2(z)))] \\ &= -E_{x \sim P_{\text{data}(x)}} \left[\log \sum_{i=1}^N w_i L_i(x) \right] \\ &\quad - E_{z \sim P_{z(z)}} \left[\log \left(\hat{x} - \sum_{i=1}^N w_i L_i(x) (G_2(z)) \right) \right]. \end{aligned} \quad (8)$$

In this article, D_i represents the i th discriminator in the multimodel discriminator, which forms the overall discriminator after weighted averaging. The proposed model improvements aim to address the challenge of generating labeled industrial control traffic samples. The specific process is as follows: noise and label data are embedded into the encoding layer, decoded by CGAN1's generator, and then evaluated by the multimodel discriminator. The resulting marginal distribution scores serve as the target scores for CGAN2, which are then trained accordingly. The detailed description of the algorithm process is shown in Algorithm 1 as follows.

IV. EXPERIMENTS AND DISCUSSION

Common industrial control datasets include ISCX2012, CIDS2017, and UNSW-NB15 [25], [26], [27]. Thomas Morris from MSU released lab-scale natural gas pipeline and water tower datasets [28]. These datasets, preprocessed to remove low-level transport data (e.g., TCP and MAC), focus on traffic between ICS control layers and SCADA. They include seven attack types targeting PLCs and normal data categories. The gas pipeline dataset has 26 features (excluding labels), and the water tower dataset has 23.

We chose these datasets because they reflect real industrial control environments, with most traffic being safe, leading to highly imbalanced label distributions. This imbalance, challenging for multiclassification, highlights the effectiveness of our sample generation. Both datasets have eight labels: seven attack types and one normal traffic type.

The datasets released by Thomas Morris have undergone certain preprocessing to strip underlying transmission data in network communications, such as TCP and MAC data. They include seven types of attacks targeting basic Internet of Things devices, such as PLCs: simple malicious response injection (NMRI), complex malicious response injection (CMRI), malicious state command injection (MSCI), malicious parameter command injection (MPCI), malicious function code injection (MFCI), denial of service (DoS), and reconnaissance, along with a normal data category. The datasets primarily focus on traffic

TABLE I
DATASET I DISTRIBUTION

Type	Label	Description	Number
Normal	0	Normal	214580
NMRI	1	Inject random response packets	7753
CMRI	2	Hide real state of controlled process	13035
MSCI	3	Inject malicious state commands	7900
MPCI	4	Inject malicious parameters commands	20412
MFCI	5	Inject malicious function code commands	4898
DoS	6	Denial of service targeting communication link	2176
Recon	7	Pretend to read from devices	3874

TABLE II
DATASET II DISTRIBUTION

Type	Label	Number
Normal	0	172415
NMRI	1	9187
CMRI	2	12460
MSCI	3	1833
MPCI	4	3725
MFCI	5	1320
DoS	6	1237
Recon	7	34002

data between the control layer of ICSs and SCADA. The natural gas pipeline dataset contains 26 features excluding labels, while the water tower dataset has 23 features excluding labels. The specific dataset classifications are given in Tables I and II.

To verify the superiority of our proposed DualCGAN-AE over other common sample generation methods, we selected four popular schemes for comparison: SMOTE, traditional GAN, WGAN-GP, and AE-GAN. These include three different GAN models, with AE-GAN serving as an ablation study for DualCGAN-AE.

In the experiment, 5000 samples of each type were generated. This number balances different attack samples well and keeps generated samples from overwhelming the real ones, ensuring data authenticity.

1) *Generate sample quality assessment:* In the mode score evaluation experiment, five different sample generation methods were tested on two datasets. The mode score calculation involved the inertia clustering evaluation index, which is the sum of squared distances from all data points to their cluster centers. Lower inertia indicates better clustering, meaning more similar samples within each cluster.

During training, cluster size parameters were tested from 1 to 9. The proposed DualCGAN-AE and SMOTE showed better inertia, while WGAN-GP had lower inertia but better mode score performance. However, SMOTE's final sample distribution mode score was poor; Table III gives the mode scores for all generation methods. Under the mode score criteria, DualCGAN-AE performed best, followed by WGAN-GP, with SMOTE performing relatively poorly.

In GAN performance evaluation, MMD measures the discrepancy between the distributions of generated and real samples by mapping them to a high-dimensional space using a kernel function and computing the mean difference. The Gauss

TABLE III
MODE SCORES UNDER TWO DATASETS

Type	Dataset I	Dataset II
SMOTE	0.61	0.72
GAN	0.99	0.98
WGAN-GP	1.07	0.96
AE-GAN	1.03	0.87
DualCGAN-AE	1.53	1.44

TABLE IV
GAUSS MMD OF VARIOUS SAMPLE GENERATION METHODS

Type	Dataset I	Dataset II
SMOTE	0.18	0.09
GAN	0.15	0.11
WGAN-GP	0.14	0.10
AE-GAN	0.27	0.26
DualCGAN-AE	0.06	0.06

TABLE V
KMMD OF VARIOUS SAMPLE GENERATION METHODS ON DATASET I

Type	Linear	Polynomial	Laplace
SMOTE	2.07×1070	1.24×10217	0.15
GAN	5.46×1068	6.02×10217	0.17
WGAN-GP	9.56×1068	2.54×10216	0.16
AE-GAN	1.60×1069	5.51×10214	0.25
AE-DGAN	2.84×1069	7.64×10216	0.10

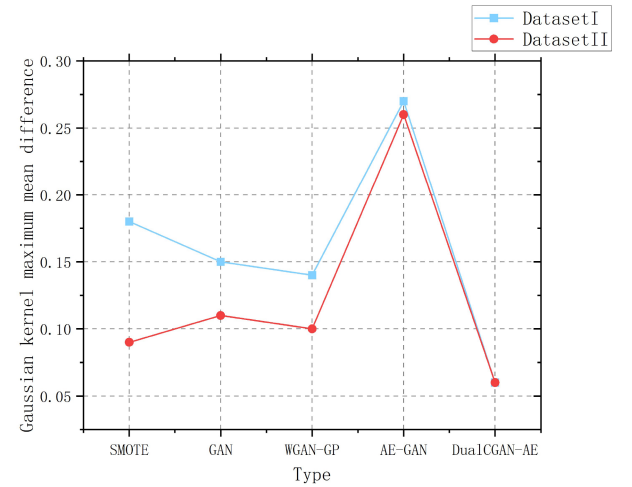


Fig. 4. Gaussian KMM D of various sample generation methods on different datasets.

MMD values for five sample generation methods across different datasets are given in Table IV, with comparisons in Fig. 4. The results indicate that DualCGAN-AE outperforms the other methods in terms of MMD, followed by WGAN-GP, while SMOTE shows relatively poorer performance.

In GAN performance evaluation, maximum mean discrepancy (MMD) measures the distribution difference between generated and real samples by mapping them to a high-dimensional space using a kernel function and computing the mean difference. Smaller MMD values indicate more similar distributions, meaning the generated samples are closer to the real ones. In this study, four kernel functions were tested, but the Gaussian kernel was chosen, as it aligns with the industrial control environment's

TABLE VI
1-NN SCORES ON TWO DATASETS

Type	Dataset I	Dataset II
SMOTE	0.72	0.83
GAN	0.82	0.90
WGAN-GP	0.80	0.88
AE-GAN	0.70	0.80
DualCGAN-AE	0.63	0.75

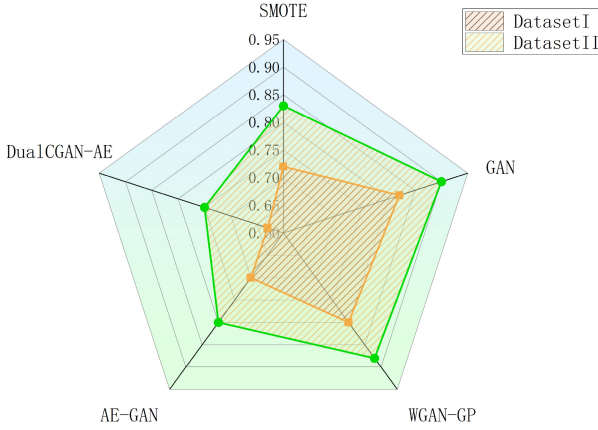


Fig. 5. Radar chart of 1-NN classification accuracy for various sample generation methods.

noise distribution. Due to large datasets, this article uses random sampling to approximate MMD calculation, effectively reducing memory requirements.

For kernel maximum mean discrepancy (KMMD), it can be used to calculate the distribution difference between generated samples and real samples in the performance evaluation of GANs. By using different kernel functions, the corresponding samples are mapped into a high-dimensional space to calculate the corresponding distribution differences for effect comparison. It can be seen that except for the Gaussian kernel function, the fluctuations of the linear kernel and polynomial kernel are relatively large, making them unsuitable as measurement standards. Although the Laplacian kernel function has a smaller fluctuation range, the Gaussian kernel function is finally selected for experimental evaluation according to the principle of choosing kernel functions based on the noise distribution characteristics of industrial control environments as much as possible. The kernel MMD value experiments are carried out for five different sample generation schemes, as given in Table V.

In addition to the above two evaluation metrics, this article employs the 1-NN classifier as a third metric for sample generation quality. Unlike traditional classification evaluations where higher 1-NN scores indicate better performance, for sample generation, mixing generated and real samples and achieving a classification accuracy close to 50% (i.e., random level) suggest better generated sample quality. Specific results are given in Table VI and Fig. 5.

In addition, when the model processes rare attack types in the experimental dataset (such as DoS and MFCI, which usually have fewer original samples), the quality of generated samples may be constrained by the scarcity of training data, thereby leading to insufficient diversity or mode collapse in generated

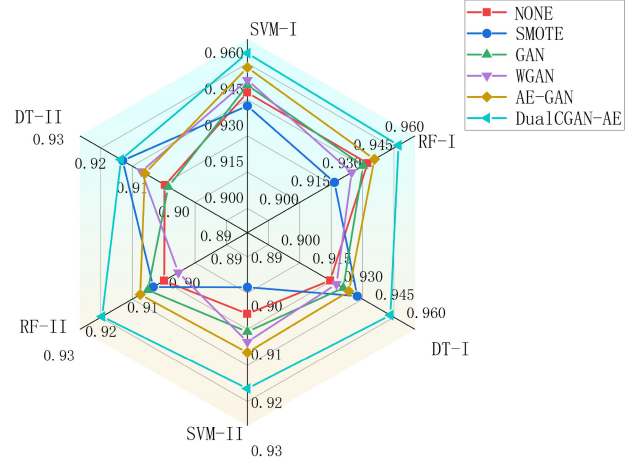


Fig. 6. Intrusion detection accuracy of various sample.

samples. Although DualCGAN-AE optimizes the generation of minority-class samples through its dual-CGAN architecture, the detection accuracy of these extremely rare categories has only increased by approximately 8% compared with baseline methods, which is still lower than that of common attack types. This indicates that when the proportion of original samples in the minority class is below a critical threshold, the model's ability to capture fine-grained features may be affected.

As shown in the 1-NN classification accuracy results, a score closer to the random distribution probability of 0.5 indicates that the generated samples are more realistic. The DualCGAN-AE model proposed in this article achieves 1-NN scores closer to 0.5 on both datasets, outperforming the other four methods. In addition, the ablation study AE-GAN also shows good performance. This highlights the importance of using multiple evaluation metrics for sample quality assessment. While AE-GAN may perform poorly in terms of Gaussian kernel mean maximum difference, it performs well in the 1-NN classifier evaluation.

2) Impact of Generated Samples on Intrusion Detection Performance: In the experiments assessing the impact of generated samples on intrusion detection algorithms, three machine learning models were used: SVM, random forest (RF), and decision tree (DT). These algorithms were chosen as they are common in intrusion detection and require minimal manual tuning, allowing a clearer assessment of sample quality on results.

To ensure result authenticity and fairness, the original datasets were split into training, validation, and test sets in an 8:1:1 ratio. We generated 5000 samples for each type of label, and after generating new samples, they were mixed with the original data to create a new dataset for retraining the three intrusion detection algorithms and calculating their accuracy. Experimental results are given in Table VII. The radar chart for the result comparison is shown in Fig. 6.

The DualCGAN-AE model enhances the performance of industrial control intrusion detection algorithms in two main ways. First, it trains on global data by leveraging the characteristics of CGAN to train labeled samples as unlabeled ones. This prevents model training bias caused by insufficient samples. Second, it can learn the long-tail distribution of samples, particularly

Algorithm 1: DualCGAN-AE Model Training Algorithm.

Input: σ , Gaussian distribution variance;
 θ_E , parameters of the initial encoder E ;
 $\theta_G^{CGAN1}, \theta_G^{CGAN2}$, parameters of the initial generator G_1
and G_2 ;
 $\theta_{D1}^{CGAN1}, \theta_{D2}^{CGAN1}, \theta_{D3}^{CGAN1}$ and
 $\theta_{D1}^{CGAN2}, \theta_{D2}^{CGAN2}, \theta_{D3}^{CGAN2}$,
parameters of the initial discriminator D_1 and D_2 ;
 w_1, w_2, w_3 are the weight of three discriminator models,
 $w_1 = w_2 = 0.4, w_3 = 0.2$;
 $\lambda_1, \lambda_2, \lambda_3$ are the learning rates of encoder, generator and
discriminator,
 $\lambda_1 = \lambda_2 = 5 \times 10^{-4}, \lambda_3 = 1 \times 10^{-3}$

WHILE $\theta_G^{CGAN1}, \theta_G^{CGAN2}$ **do not reach convergence:**

/*Step 1: Data compression processing:*/

$(x, y) = \text{Batch}(\text{Trainingset})$ //Get Label

Calc AE: $z = E((x, y); \theta_E)$

/*Step 2: Training CGAN1: */

$x'_1 = G_1(z; \theta_G^{CGAN1})$

$L_E = \frac{1}{N} \sum_{i=1}^N (G_1(x) - E(x'_1))^2$

Update encoder parameters: θ_E

$D_{CGAN1} = w_1 D_{D1}^{CGAN1}(x'_1; \theta_{D1}^{CGAN1}) +$
 $w_2 D_{D2}^{CGAN1}(x'_1; \theta_{D2}^{CGAN1}) + w_3 D_{D3}^{CGAN1}(x'_1; \theta_{D3}^{CGAN1})$

$\hat{x} = \text{average}(\text{Traniningset}, 2\sigma)$

$L_D^{CGAN1} = -E_{x \sim P_{data}(x)} [\log D_1(x)]$

$-E_{z \sim P_z(z)} [\log(1 - D_1(G_1(z)))]$

Update discriminator parameters:

$\theta_{D1}^{CGAN1}, \theta_{D2}^{CGAN1}, \theta_{D3}^{CGAN1}$
 $L_G^{CGAN1} = w_{enc} \frac{1}{N} \sum_{i=1}^N (G_E(x) - E(x'))^2$
 $+ w_{adv} E_{z \sim P_z(z)} [\log(1 - D_1(G_1(z)))]$

Update generator parameters: θ_G^{CGAN1}

/*Step 3: Training CGAN2: */

$x'_2 = G_2(z; \theta_G^{CGAN2})$

$L_E = \frac{1}{N} \sum_{i=1}^N (G_2(x) - E(x'_2))^2$

Update encoder parameters: θ_E

$D_{CGAN2} = w_1 D_{D1}^{CGAN2}(x'_2; \theta_{D1}^{CGAN2}) +$
 $w_2 D_{D2}^{CGAN2}(x'_2; \theta_{D2}^{CGAN2}) + w_3 D_{D3}^{CGAN2}(x'_2; \theta_{D3}^{CGAN2})$

$L_D^{CGAN2} = -E_{x \sim P_{data}(x)} [\log D_2(x)]$

$-E_{z \sim P_z(z)} [\log(\hat{x} - D_2(G_2(z)))]$

Update discriminator parameters:

$\theta_{D1}^{CGAN2}, \theta_{D2}^{CGAN2}, \theta_{D3}^{CGAN2}$
 $L_G^{CGAN2} = w_{enc} \frac{1}{N} \sum_{i=1}^N (G_E(x) - E(x'))^2$
 $+ w_{adv} E_{z \sim P_z(z)} [\log(\hat{x} - D_2(G_2(z)))]$

Update generator parameter: θ_G^{CGAN2}

END WHILE

TABLE VII

INTRUSION DETECTION ACCURACY OF MACHINE LEARNING ALGORITHMS ON DATASETS

Type	None	SMOTE	GAN	AE-GAN	DualCGAN-AE
SVM1	94.79%	94.24%	95.10%	95.83%	96.42%
RF1	94.74%	93.16%	94.53%	95.07%	96.22%
DT1	92.96%	94.25%	93.58%	93.85%	95.83%
SVMII	90.68%	90.13%	91.05%	91.48%	92.23%
RFII	90.99%	91.25%	91.38%	91.56%	92.47%
DTII	90.97%	91.98%	90.89%	91.45%	92.02%

TABLE VIII

INTRUSION DETECTION ACCURACY OF DEEP LEARNING ALGORITHMS ON DATASET I

Type	None	SMOTE	GAN	AE-GAN	DualCGAN-AE
ICNN	95.40%	95.14%	95.65%	95.34%	95.80%
GoogelNet	95.60%	95.37%	95.68%	95.75%	95.92%
BiLSTM	95.45%	95.51%	95.70%	95.93%	95.16%
PSO-GRU	95.35%	95.40%	95.03%	94.89%	95.44%

focusing on the easily overlooked marginal distribution features. It generates samples in proportion to ensure the generated samples retain the long-tail distribution characteristics of industrial control traffic. This guarantees distribution consistency in the mixed samples. As a result, when intrusion detection algorithms are trained on these mixed samples, they learn the distribution features of industrial control traffic more comprehensively, rather than just focusing on centrally distributed data.

Table VIII gives the intrusion detection accuracy of deep learning algorithms on Dataset I. From the table, we can see that the DualCGAN-AE model demonstrates distinct performance patterns when applied to deep learning and machine learning methods. In terms of accuracy, the mixed samples generated by DualCGAN-AE proposed in this article provide more positive effects on industrial control intrusion detection algorithms compared with other schemes. The quality of samples generated by other schemes varies: although they can improve the final results in some detection algorithms, there are still cases where the final results may decrease. The time-series schemes generally show better results than the convolutional schemes, which is related to the fact that industrial control traffic data have fewer dimensional characteristics and the spatial extraction effect is not good.

In summary, the DualCGAN-AE model can generate traffic data that better match the distribution characteristics of ICSs. The generated samples are diverse and similar to real data, effectively improving the accuracy of intrusion detection algorithms and unlocking their potential for performance improvement.

V. CONCLUSION

In recent years, with the expansion of ICSs, cybersecurity has become a critical issue. Both upper layer control hosts and lower layer industrial devices rely on traffic for information transmission, where the traffic may contain malicious instructions, posing significant security risks to industrial systems. These systems use traffic to transmit information between control devices and industrial equipment but they may be exploited by malicious actors to disrupt operations. To address the above problems and improve the actual detection accuracy of intrusion detection in ICSs, this article focuses on enhancing the application of GANs in sample generation. We propose the DualCGAN-AE model, which improves the model structure, training algorithm, and intrusion detection model for ICS traffic samples. This model strengthens training stability and the ability to learn marginal distribution data. Experiments conducted on two datasets, evaluating both data quality and impacts on intrusion detection

performance, demonstrate that the proposed DualCGAN-AE solution significantly improves detection accuracy and has the potential for further performance enhancements.

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